

Algebra 1
Unit 2
Lesson 4 Homework

Name Answer Key
Date _____
Period _____

1. True or False? The polynomial $-7xy^4 + 4x^5y^2 - 8.6x^3y^3$ is written in standard form.

False

2. Write the polynomial $-7xy^4 + 4x^5y^2 - 8.6x^3y^3$ in standard form.

$$4x^5y^2 - 8.6x^3y^3 - 7xy^4$$

For questions 3-4, rewrite each polynomial in standard form, then identify the features of the polynomial.

3. $2.5y^3 + 7y - 8\frac{7}{8}y^5$

Standard Form: $-8\frac{7}{8}y^5 + 2.5y^3 + 7y$

Number of Terms: 3

Degree of the polynomial: 5

Leading Term: $-8\frac{7}{8}y^5$

Leading Coefficient: $-8\frac{7}{8}$

4. $-\frac{3}{4}a^4b^2 - 16ab^3 + ab^6$

Standard Form: $ab^6 - \frac{3}{4}a^4b^2 - 16ab^3$

Number of Terms: 3

Degree of the polynomial: 7

Leading Term: ab^6

Leading Coefficient: 1

5. Create a polynomial that has 4 terms, a degree of 6, a leading coefficient of -1 , and a constant term of -5.5 .

$$-x^6 + 2x^5 - 3x^2 - 5.5 \quad \text{Answers will vary.}$$

6. Samantha found the product of $-2g^3(\frac{1}{4}g^8h^3 + 5g^2h^6 - g^9)$ to be $-\frac{1}{2}g^{24}h^3 - 10g^6h^6 - 2g^{27}$. Do you agree or disagree with her? Explain.

Disagree. She did not apply the law of exponents correctly.

7. Find the product of $-2g^3(\frac{1}{4}g^8h^3 + 5g^2h^6 - g^9)$ in standard form.

$$-\frac{1}{2}g^{11}h^3 - 10g^5h^6 + 2g^{12}$$

$$-\frac{1}{2}g^{11}h^3 + 2g^{12} - 10g^5h^6$$

Find the product of each by distributing the monomial to the polynomial. Express the product in standard form.

8. $-5x^2(-3xy + 2x^7 - 7x^3)$

$$15x^3y - 10x^9 + 35x^5$$

$$-10x^9 + 35x^5 + 15x^3y$$

9. $(-\frac{5}{8}c^4d + 10c^5d^3 - cd^2)(\frac{3}{5}c^2)$

$$-\frac{3}{8}c^6d + 6c^7d^3 - \frac{3}{5}c^3d^2$$

$$6c^7d^3 - \frac{3}{8}c^6d - \frac{3}{5}c^3d^2$$

10. $-4.2a^3(2b^5 - 1.5a^2b - 9)$

$$-8.4a^3b^5 + 6.3a^5b + 37.8a^3$$